Futures Trading and Spot Market Volatility: A Case of S&P CNX Nifty Index

P. Sakthivel[*] and B. Kamaiah[**]

Abstract

An attempt is made here to investigate the relationship between stock market volatility and trading activity (trading volume and open interest) in Nifty futures market using the GARCH framework. The study uses daily closing price of Nifty and trading volume, and open interest for Nifty index futures for the period July 1, 2000 to February 28, 2008. The study considers only near-month contract data since most trading activities taken place in this contract. Open interest and volume are used as measures of futures trading activity. The study decomposes series of futures trading volume and open interest into expected and unexpected components by an appropriate ARMA procedure. Then, the study introduces futures trading variables such as expected and unexpected futures volume and open interest in GARCH and GJR GARCH models to examine their effects on spot market volatility. The results show that the spot market volatility bears a positive relationship with unexpected trading volume and open interest in Nifty futures market. But, the same bears a negative relationship with expected futures trading volume and open interest.

Keywords: Futures Trading, GJR GARCH and Volatility

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I. Introduction

The price (volatility)-volume relation in financial markets has received considerable attention in the literature. It provides an insight into the structure of financial markets, revealing how information flows into the market, how the same is disseminated and the extent to which market prices convey the information. This relationship can be used to examine the usefulness of technical analysis, and has important implications for fashioning new contracts.

Broadly there are two strands of explanations for the price-volume relationship namely, information theories, such as mixture of distribution theory, and sequential arrival of information theory. Both the theories relate to the information arrival process. The mixture of distribution theory states that the price-volume relation originates from a joint dependence on a common event, often interpreted as rate of information flow to the market (Clerk, 1973; Epps and Epps, 1976; Lawrence and Harris, 1986). According to this theory, price volatility and trading volume are positively and contemporaneously correlated. An alternative explanation has been given by the sequential arrival of information theory [Copeland (1976), Morse (1981), Jennings and Barry (1983)]. According to this theory, new information is disseminated sequentially to the traders, and investors receive information in groups, across time and trading occurs after each group's reception of information.

For mixture of distribution theory, since all traders receive information simultaneously, there should be no information in past observations of volatility that can be used to forecast future volume that is not already contained in the lagged values of volume and similarly for forecasting volatility. The sequential arrival of information theory is consistent with a dynamic relationship between variables, when past volume provides information on current volatility and past volatility contains information about current volume. Therefore, despite the fact that both theories imply a positive relationship between price volatility and trade volume, the predicated dynamic response pattern is

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expected to be different. The sequential arrival of information theory implies a lead–lag relation in both the directions between volume and volatility, while mixture of distribution theory suggests that a positive contemporaneous causal relationship between the two variables.

Though there are quite a few studies in the literature focusing on the price volatility – volume relationship, they are largely confined to the developed markets. In the context of emerging economies like India, there is no study highlighting this relationship. Hence the present study is undertaken. Having adopted the GARCH framework of analysis, the study as such is not intended to verify any of the theories mentioned above.

II. Futures Trading Activity and Spot Market Volatility: Some Empirical Evidences:

There are studies which have investigated the relationship between stock market volatility and open interest and trading volume in futures market. Some researchers (Figlewski, 1981; Bessembinder and Seguin, (1992,1993); Darrat and Rahman, 1995; Chen et al, 1995; Chatrath et al, 1996; Adrangi and Chatrath, 1998; and Gulen and Mayhew, 2000; Jeanneau, 2003) investigated related but different question: how does the level of futures trading activity (volume and/or open interest) affect cash market volatility? Figlewski (1981) found a positive contemporaneous association between GNMA cash price volatility and futures trading activity (both trading volume and open interest). Similarly, Bessembinder and Seguin (1992) found that information flow, measured by trading volume, had a positive relationship with volatility. They concluded that market depth, measured by open interest, had an inverse relationship with volatility, as market depth increased, volatility decreased.

Chen, *et al* (1995) documented a positive contemporaneous association between stock index volatility and unexpected components of futures trading activity. Jeanneau (2003) argued that speculative or information based transactions can create a link between spot market volatility and trading activity in derivatives market. This link depends in part on whether the new information is privative or public and on the type of asset traded. The arrival of new information first reflected in futures market leading to unexpected futures volume, a positive relationship between volatility and unexpected futures trading activity.

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Further, researchers extended this line of work by applying causality tests and found mixed results. Darrat and Rahman (1995) reported no evidence of causality running from S&P 500 futures trading activity (both volume and open interest) to cash price volatility. In contrast, Chatrath, et al (1996) argued that currency futures trading activity (trading volume) had a significant positive (i.e., destabilizing) causal impact on cash price volatility, with a weaker negative causal influence from spot exchange rate volatility to futures trading activity. Adrangi and Chatrath (1998) reported no causality between cash market volatility and open interest position of large hedgers. They also found that an increase in the unexpected component of open interest positions of large speculators and small traders caused higher cash price volatility, which was interpreted as evidence of a destabilizing effect.

In contrast, Smith (1989) found that S & P 500 futures volume had no effect on changes in the volatility of S & P 500 index returns. Santoni (1987) found a negative correlation between S & P futures volume and daily spot close, suggesting an increase in futures trading does not lead to an increase in volatility of index. Dennis and Sim (1999) found that FTSE 100 futures volume had no significant impact on spot market volatility. Further, a study of 25 countries by Gulen and Mayhew (2000) found that futures volume had a positive effect on spot market volatility in Denmark, Germany and Hong Kong, and a negative effect in Austria and U.K and no effect in remaining 18 countries.

Lanmoureux and Lastrapes (1990) were the first to apply stochastic time series models of conditional heteroscedasticity (GARCH-type) in the context of price-volume investigations. They found that the persistence of stock return variance vanishes when trading volume is included in the conditional variance equation. Considering that trading volume is a proxy for the flow of information into the market, they found that trading volume and return volatility are driven by the same factors. They do not, however, answer the question on the identity of these factors.

The effects of futures trading activity on spot market volatility was examined by Bessembinder and Seguin (1993). They argued that there was no evidence of futures' trading activity (volume and open interest) destabilizing the spot market. They employed ARIMA model to decompose series of futures trading volume and open interest into expected and unexpected components. The study also employed unbiased standard deviation model to examine relationship between futures trading volume and stock market volatility. They showed that equity volatility co-varies positively with unexpected futures volume, but forecastable futures trading activity. Further, though futures trading

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is systematically related to the futures contract cycle, they found no evidence of relationship between futures life cycle and spot equity volatility.

Bhar and Malliaries (1998) investigated trading volume, price, short and long-term relationships between price and volume and the determinants of trading volume in foreign currency futures. They used five currency futures namely, British Pound, Canadian Dollar, Japanese Yen, German Mark and Swiss Francs. They collected data nearby futures contracts period from May 1972 to November 1994. They found that all the five currencies that the price volatility is a determinant of the trading volume changes. Further, the price volatility is a determinant of the unexpected volume changes. Also, there is a significant relationship between the volatility of price and the volatility of trading volume changes for three of the five currencies in the daily frequency and for only one currency in the monthly frequency.

Gulen and Mayhew (2000) investigated the effects of futures trading activity on international equity market volatility. The study decomposed series of futures trading volume and open interest into expected and unexpected components by an appropriate ARMA framework. Then the study included futures trading variables such as expected and unexpected futures volume and open interest in GARCH and examined their effects on spot price volatility. The study suggested that futures trading activity (volume and open interest) destabilizes equity market. The study found that in most countries, volatility tends to lower in periods when open interest in stock index futures is high. In some cases, it was found that volatility is higher in periods when futures volume is high, but this is driven by the unexpected component of volume not the expected component.

Griffin, *et al* (2004) investigated the dynamic relation between past returns and trading activity for 46 stock markets, finding a strong positive relation between past returns and turnover. Furthermore, they document evidence of a stronger relation between return and turnover in countries with restrictions on short sales, where corruption is higher, and where the allocative efficiency of the stock market is weaker (developing markets). The lagged return-volume relation is insignificant for the U.S. markets between 1993 and 2003. The relation is also stronger among individual investors than among foreign or institutional investors. These results are consistent with Karpoff (1987), who observed the effect of short sales on this relation for the stock market.

The relationship between speculative futures trading activity and bond market volatility was investigated by Kim (2005). In the context of Korea, and applying VAR model, he found a positive relationship between bond market volatility and speculative ratio. The

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results imply that new variable could be good candidate, reflecting the speculative trading activities' in derivative market.

Mahmood and Salleh (2006), examined the relations between returns, trade volume and market depth for two futures contracts namely, Stock Index (SI) futures and Crude Palm Oil (CPO) futures traded at the Kuala Lumpur Option and Financial Futures (KLOFFE), and Commodity and Monetary Exchange (COMMEX), respectively. They used daily settlement prices, trading volumes and open interest for Crude Palm Oil Futures and Stock Index Futures period from December 15, 1995 to January 19, 2001. They considered futures volume as well as open interest, a proxy of market depth, on volatility. Since both volume and open interest are highly serially correlated, these variables are divided into expected and unexpected components using ARIMA framework. They found that unexpected volume and open interest are more likely to have a greater impact on volatility than the expected volume and open interest. They concluded that unexpected volume and market depth proxies by unexpected open interest do have information which can have an effect on return volatility for Malaysian futures markets.

From the foregoing review, it becomes clear that studies examining the relationship between stock market volatility and futures trading activity have been inconclusive. The relationship needs to be probed into further. In the Indian context, there are no comprehensive studies focusing on this issue. Hence, the present study seeks to empirically investigate whether futures trading activity (trading volume and open interest) affects spot market volatility or not.

III. Data and Methodology

The data for present study are obtained from the official web page of NSE i.e.www.nesindia.com. The daily closing price of Nifty and trading volume, and open interest for Nifty index futures are collected from July 1, 2000 to February 28, 2008. The study considered only near-month contract data since most trading activities taken place in this contract. Open interest and volume are used as measures of futures trading activity. Open interest refers to the total number of contracts (either short or long) open on any particular day. If a new buyer (a long) and new seller (a short) enter a trade, open interest increases by one. However, if a trader already holding a long position sells to a new trader wanting to initiate a long position open interest remains the same. And if a trader holding a long position sells to a trader wanting to get free of his existing short position, open interest decreases by one.

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Construction of Proxies for the Futures Trading Activity

Futures volume and open interest may be decomposed into expected and unexpected components, to represent information less trading activity and information arrival respectively. Unexpected trading volume is said to occur when there is a sudden change in trading volume within day due to information shocks, while expected component can be considered as the natural activity in futures market, that is, futures trading volume, when there is no relevant new information arrival at market.

In this study, series of futures volume and open interest are decomposed into expected and unexpected components by an ARMA procedure. This is done for the futures volume and open interest separately and all the four variables are included as exogenous explanatory variables in GARCH and GJR GARCH models. Inclusion of volume and open interest as explanatory variable in the conditional variance equation would lead to the problem of multi-colinearity. To avoid this problem, futures trading activity is partitioned into expected and unexpected (shocks) components using ARMA (5, 5) model. After estimation of the model, the residuals are taken to proxy of unexpected futures volume and open interest, while expected component is defined as difference between actual and the unexpected compo series.

Methodology

Autoregressive Conditional Heteroskedasticity (ARCH) has been identified as common occurrence in financial markets data. The persistence of variance after price shocks has become an important issue in the examination of the volume- volatility relation. An explanation for the presence of ARCH effects is that daily returns are generated by a mixture of distributions, where the rate of daily information arrival is the stochastic mixing variable. The ARCH effect might capture the entire time series properties of the information mixing variable. Lamoureux and Lastapes (1990) showed that ARCH follows from the series correlation of the mixing variables – number of price changes (which represent the number of information arrival).

The present study found GARCH specification more appropriate than the standard statistical models because it is consistent with a return distribution, which is leptokurtic, and it also allows for a long-term memory in the variance of the conditional return distributions. GARCH model is capable of mimicking observed statistical characteristics of many time series of return on financial assets.

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The GARCH analysis might be useful for the purpose of examining the relationship between volume and price volatility. The introduction of volume into the GARCH equation is more a test of the volume-volatility relationship. The GARCH (1,1) regression model is obtained by:

$$R_t = \Phi_0 + \Phi_1 R_{t-1} + \varepsilon_t \quad \ldots\ldots (1)$$

$$\varepsilon_t \mid \Psi_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} \quad \ldots\ldots (2)$$

where R_t is log return conditional on past information, which is proxy by R_{t-1} and α_0, α_1 and β_1 are the parameters to be estimated. Ψ_{t-1} is the information set time t_{-1}, ε_t is the stochastic error conditional on Ψ_{t-1} and is assumed to normally distribution with zero mean and conditional (time varying) variance.

Since the present study proposes to investigate the relationship between stock market volatility and trading activity (volume and open interest) in Nifty futures market, futures trading variables such as volume and open interest are introduced in GARCH conditional equation. Specifically the incremental contribution of expected end unexpected futures volume and open interest on Nifty index volatility are estimated. Formally,

$$R_t = \Phi_0 + \Phi_1 R_{t-1} + \varepsilon_t$$

$$\varepsilon_t \mid \Psi_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} + \delta_1 Exvol + \delta_2 Unexvol + \delta_3 Exoi + \delta_4 Unexoi \quad \ldots\ldots (3)$$

where Exvol and Exoi are expected futures volume and open interest respectively. Similarly, Unexvol and Unexoi are unexpected futures volume and open interest respectively.

The GARCH model cannot capture the asymmetry response of volatility to news. To accommodate for the asymmetry, GJR-GARCH (1, 1) model is estimated by including expected and unexpected component of volume and open interest in the conditional variance equation. The model is specified as:

$$R_t = \Phi_0 + \Phi_1 R_{t-1} + \varepsilon_t$$

$$\varepsilon_t \mid \Psi_{t-1} \sim N(0, h_t)$$

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$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} + \gamma_1 I_{t-1} \varepsilon_{t-1}^2 + \delta_1 Exvol + \delta_2 Unexvol + \delta_3 Exoi + \delta_4 Unexoi \quad \text{.... (4)}$$

Where $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$,

$= 0$ otherwise

IV. Empirical Analysis:

The empirical results are presented in tables 1 to 3. Table 1 presents a wide range of summary statistics on Nifty index, volume and open interest of Nifty futures market for the full sample period that is July 1, 2000 to February 28, 2008. Basic statistics used in the study are described. These include the mean, standard deviation, skewness, kurtosis, maximum and minimum. The mean daily percentage return on the Nifty index is . 0.000315 % with a standard deviation of 0.0856. The negative sample skewness coefficient i.e. -0.2789 indicates that the frequency distribution of the return series is negatively skewed or longer tail to the left. The value of kurtosis (7.486) shows that the unconditional distribution of return exhibits fat tails and excess peaked ness at the mean than the corresponding normal distributions.

The non-normality can be conformed by the Jarque-Bera (JB) test. The JB statistic is highly statistically significant and hence we reject the null hypothesis. The result of LB-Q test also rejects the null hypothesis of identical and independent. Hence, the significant autocorrelation in returns series proves the presence of volatility clustering. The LM test statistic shows that there is evidence for ARCH effect and time varying volatility. Hence, the preliminary data analysis of the Nifty index returns confirms that there is time varying volatility, fat tail distribution and volatility clusters.

Table 1 also shows that mean of daily volume is 0.0045 percent and open interest is 0.0409 in Nifty futures market. The skewness of open interest on Nifty futures market is measured at 1.4230, suggesting that open interest follows heavy tail distribution with right side. However, futures volume exhibit negative skewness in distribution with heavier tails to left. The kurtosis value exceed more than 3 for futures volume and open interest indicates that these two series follow non normal distribution. Logarithm of trading volume and open interest series has been taken for the analysis. Both the series are found to be stationary and non-normal.

Table 2 provides result of GARCH model with expected and unexpected components. The results show that estimated coefficient of unexpected trading futures volume is positive

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(0.04035) and significant which indicates that there is a positive relationship between spot market volatility and unexpected trading volume in Nifty futures market. In other words, unexpected futures trading volume has a positive significant effect on spot market volatility. This may be because of the fact that new information disseminates faster in futures market than in the spot market and consistent with the hypothesis that information stocks generate unexpected trading volume in Nifty futures market and the unexpected futures volume shock might influence the spot market volatility. However, the estimated coefficient of expected futures trading volume is found to be negative (-3.483) implying that there is negative relationship between stock market volatility and expected trading volume in Nifty futures market or expected futures trading volume has a negative significant effect on spot market volatility. The results show that the expected futures volume does not contribute to explaining the conditional spot market volatility, because there is no relevant new information arrival into market.

The estimated coefficient of expected open interest is significant at 1% level for Nifty index futures market. However, when the study replaces the expected with unexpected open interest into conditional variance equation, the estimated coefficient of unexpected open interest turned out be positive (0.1540) implying that there is positive relationship between stock market volatility and unexpected open interest. In other words, unexpected open interest has positive and significant effects on spot market volatility. But estimated coefficient of expected open interest is negative (-2.0921) indicating that that there is negative relationship between stock market volatility and expected open interest in Nifty futures market.

The results of GJR GARCH model with expected and unexpected components are given in table 3. The asymmetric parameter is 0.745 which is positive and highly statistically significant, implies leverage effect in volatility. By comparing GARCH and augmented GJR-GARCH with volume and open interest, we found that the degree of persistent measured by the sum of ARCH and GARCH coefficient has reduced from 0.94 to 0.75. The coefficients of unexpected volume and open interest are statistically significant and positive. However, the coefficient of expected futures volume and open interest are negative and statistically significant. But the coefficient estimates for unexpected volume is larger then expected volume, demonstrating that surprise (unexpected) in volume convey more information and thus, have a larger effect on return volatility than expected volume.

V. Concluding Remarks:

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The study is conducted in the GARCH framework. GARCH and GJR GARCH models are used to investigate the relationship between stock market volatility and trading activity (trading volume and open interest) in Nifty futures market. After decomposing series of futures trading volume and open interest into expected and unexpected components by an appropriate ARMA procedure, they are introduced in the GARCH and GJR GARCH models to investigate their effects on the spot market volatility. The empirical results reveal that unexpected futures trading volume and open interest have positive significant effects on spot market volatility. This is due to fact that arrival of new information generates trading volume in futures as well as in spot market. A positive relationship between price volatility and trade volume may means that new futures contract will be successful only to the extent that there is enough price uncertainty associated with the underlying asset. The study also showed that expected volume and open interest have negative significant effects on spot market volatility.

The results of this study should be useful to exchanges, regulatory agencies who watch the behavior of volume of trading. Traders themselves should pay attention to trading volume. Low volume usually indicates that the market is illiquid and the bid/ask spread would tend to be large, resulting in high price volatility. Such a market would discourage hedgers, but may benefit speculators. On the other hand, high trading volume contributes to high liquidity and the bid/ask spread would tend to be low, resulting in low price variability. Hedgers prefer low day to day volatility while speculators usually do not, because low volatility reduces speculators' profits. Therefore, it is important to investigate the relationship between spot market volatility and futures trading volume.

References

Adrangi, B. and A. Chatrath (1998), "Futures Commitments and Exchange Rate Volatility", Journal of Business Finance and Accounting, Vol. 25, pp. 501–20.

Electronic copy available at: https://ssrn.com/abstract=1589962

Bessembinder, H. and P.J. Seguin (1992), "Futures Trading Activity and Stock Price Volatility", Journal of Finance, Vol. 47, pp. 2015–34.

Bessembinder, H. and P.J. Seguin (1993), "Price Volatility, Trading Volume, and Market Depth: Evidence from Futures Markets", Journal of Financial and Quantitative Analysis, Vol. 28, pp. 21–39.

Clark P (1973), "A Subordinated Stochastic Process Model with Finite Variances for Speculative Prices", Econometrica, Vol. 41, pp. 135-155.

Copeland T E (1976), "A Model of Asset Trading under the Assumption of Sequential Information Arrival", Journal of Finance, Vol. 34, pp. 1149-1168.

Chatrath, A., S. Ramchander and F. Song (1996), "The Role of Futures Trading Activity in Exchange Rate Volatility", Journal of Futures Markets, Vol. 16, pp. 561–84.

Chen, N., C.J. Cuny and R. A. Haugen (1995), "Stock Volatility and the Levels of the Basis and Open Interest in Futures Contracts", Journal of Finance, Vol. 50, pp. 281–300.

Darrat, A.F. and S. Rahman (1995), "Has Futures Trading Activity Caused Stock Price Volatility?", Journal of Futures Markets, Vol. 15, pp. 537–57.

Dennis S.A and A.B Sim (1999), " Share Price Volatility with the Introduction of Individual Share Futures on Sydney Futures Exchange", International Review of Financial Analysis, Vol.8, pp.153-163.

Epps, T. W. and M. L. Epps, (1976), "The Stochastic Dependence of Security Price Changes and Transaction Volumes: Implication for the Mixture-of-Distributions Hypothesis", Econometrica, Vol 44, 305-322.

Figlewski, S. (1981), "Futures Trading and Volatility in the GNMA Market", Journal of Finance, Vol. 36, pp. 445–56.

Griffin, J., F. Nardiri, and R. M. Stulz (2004), "Do Investors Trade More when Stocks have Performed Well? Evidence from 46 Countries", EFA 2004 Maastricht Meetings Paper No. 5211; Dice Center Working Paper No. 2004-13.

Gulen, H. and S. Mayhew (2000), "Stock Index Futures Trading and Volatility in International Equity Markets', Journal of Futures Markets", Vol. 20, pp. 661–85.

Harris and Lawrence, (1986), Cross-security tests of the Mixture of Distributions Hypothesis, Journal of Financial and Quantitative Analysis, 21, 39–46.

Jennings, R. H., and C. Barry. (1983) "Information Dissemination and Portfolio Choice." Journal of Financia1 and Quantitative Analysis, 18, 1-19.

Jeanneau S. (2000), "Volatility and Derivatives Turnover: a Tenuous Relationship", BIS Quarterly Review, pp.57-65.

Karpoff, J. (1987), "The Relation Between Price Changes and Trading Volume: A Survey", Journal of Financial and Quantitative Analysis, Vol. 22, pp. 109–26.

Kim, Joocheol (2005), "An Investigation of the Relationship between Bond Market Volatility and Trading Activities: Korea Treasury Bond Futures Market", Applied Economics Letters, Vol.12, pp.657 – 661.

Electronic copy available at: https://ssrn.com/abstract=1589962

Lanmoureux G.G and W.D Lastrapes (1990), "Heteroskedasicity in Stock Return Data: Volume Versus GARCH Effects", Journal of Finance, Vol.45, pp.221-29.

Mansor Mahmood and Aisyah Salleh (2006), "The Dynamic Relationship between Price Volatility, Trading Volume and Market Depth: Empirical Evidence from the Malaysian Futures Market", The Icfai Journal of Derivatives Markets, Vol. III, No. 4, 2006.

Morse. D. (198I), "Price and Trading Volume Reaction Surrounding Earnings Announcements: A Closer Examination", Journal of Accounting Research, 19 374-383.

Ramaprasad Bhar and Malliaries (1998), "Volume and Volatility in Foreign Currency Futures Market", Review of Quantitative Finance & Accounting, Vol.10, pp.285-302.

Santoni (1987), "Has Programmed Trading Made Stock Price Volatility", Federal Reserve Bank of St.Louis Review, pp.19-29.

Smith (1989),"Market Volatility: Causes and Consequences", Cornell Review, Vol.74, pp.953-62.

Table 1: The Summary of Descriptive Statistics on Nifty, Futures Volume and Open Interest

Statistic	Nifty Index	Open Interest	Futures

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			Volume
Mean	0.000315	0.0409	0.00458
Median	0.000776	-0.0059	-0.00491
Maximum	4.786581	3.15115	4.12901
Minimum	-7.908761	-3.67473	-4.34623
Std.dev	0.085641	0.27186	0.42389
Skew ness	-0.27892	1.42304	-0.01437
Kurtosis	7.486850	42.26615	16.9553
Jorq -Bera	2829.21	113018.53	14201.55
Residual diagnostics	Test statistic	Test statistic	Test statistic
LB-Q(12)	592.253 (0.000)	64.337 (0.000)	200.373 (0.000)
LB^2-Q(12)	73.253 (0.000)	161.094 (0.000)	165.66 (0.000)
LM(6)	370.631 (0.000)	165.664 (0.000)	226.731 (0.000)

Note: (LB-Q (k) and LB2 -Q (k) are the Ljung - Box Q test statistics for testing joint significance of autocorrelation of standardized residuals up to lags of 12. LM is the ARCH effect in the standardized squared residuals up to lags of 6. The figures in parentheses are p- values

Table 2: Estimates of GARCH (1, 1) with Expected and Unexpected Components of Futures Trading Volume and Open Interest

Parameters	Estimated Value	t- Statistic
Φ_0	0.0092	0.129
Φ_1	0.0791	1.570
α_0	0.0697	2.381*
α_1	0.3878	8.176*
β_1	0.5659	33.539*
δ_1	-0.3486	-2.911*
δ_2	0.0403	5.331*
δ_3	-2.0925	-11.372*
δ_4	0.1540	3.297*

Note: *Indicates 1% significance level. The parameters refer to the model given equations (1) to (3)

Table 3: Estimates of GJR-GARCH (1, 1) with Expected and Unexpected Components of Futures Trading Volume and Open Interest

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Parameters	Estimated value	t- Statistic
Φ_0	0.00928	0.128
Φ_1	0.079	1.570
α_0	0.002	5.645*
α_1	0.150	2.148*
β_1	0.601	8.163*
γ_1	0.745	2.557*
δ_1	-0.826	-3.227*
δ_2	1.153	4.784*
δ_3	-1.762	-2.315*
δ_4	0.763	3.784*

Note: *Indicates 1% significance level. The parameters refer to the model given equations (1) and (4)

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